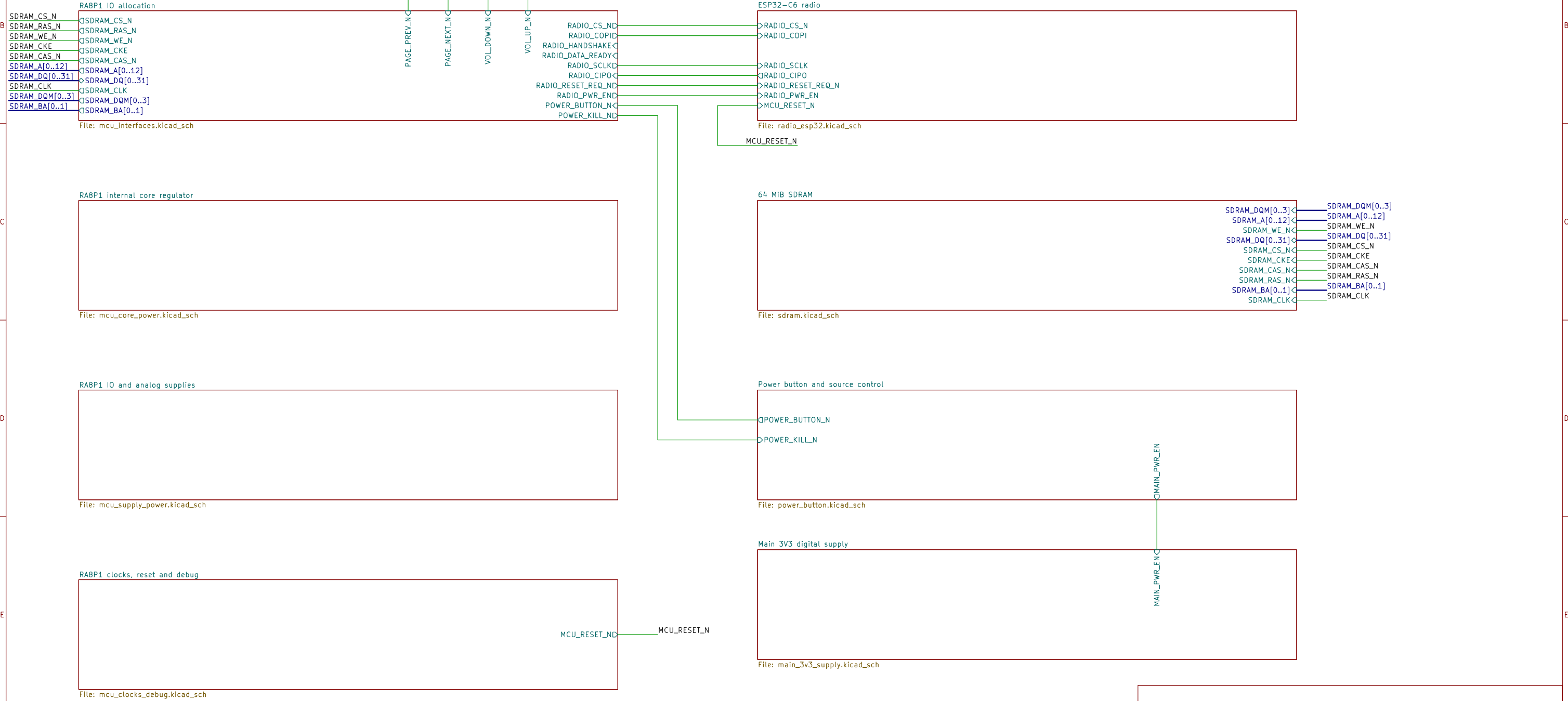


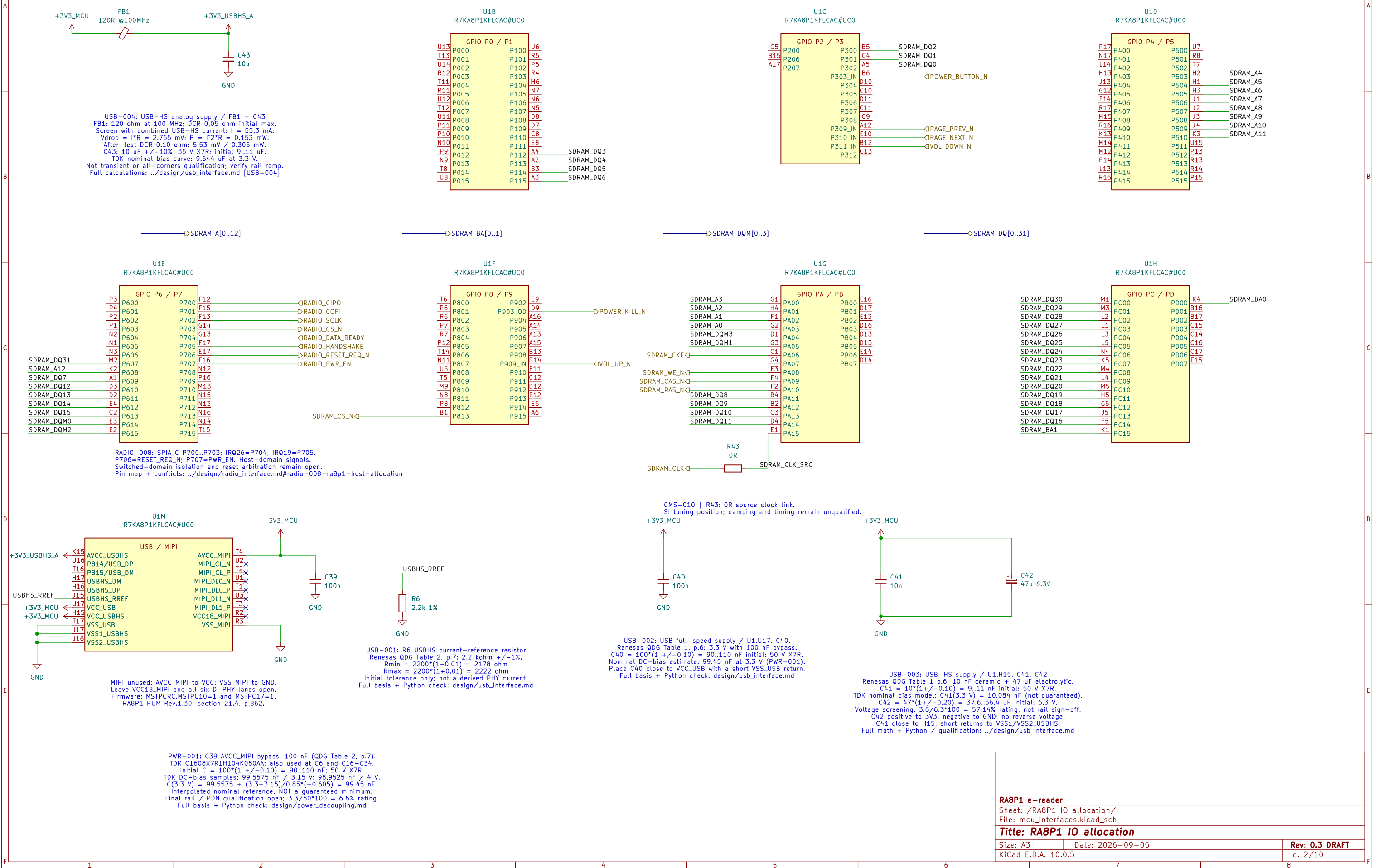
RA8P1 E-READER

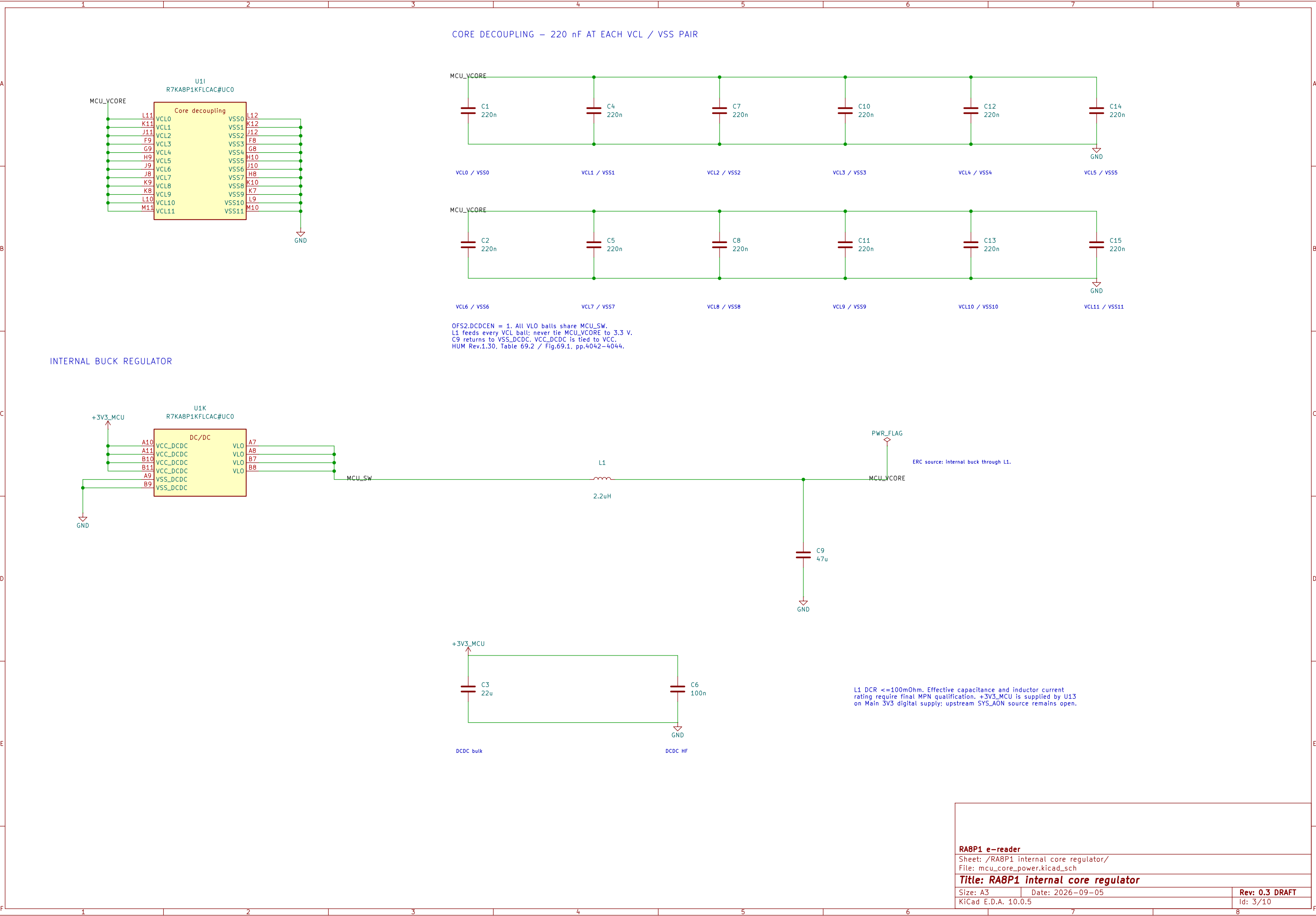
Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

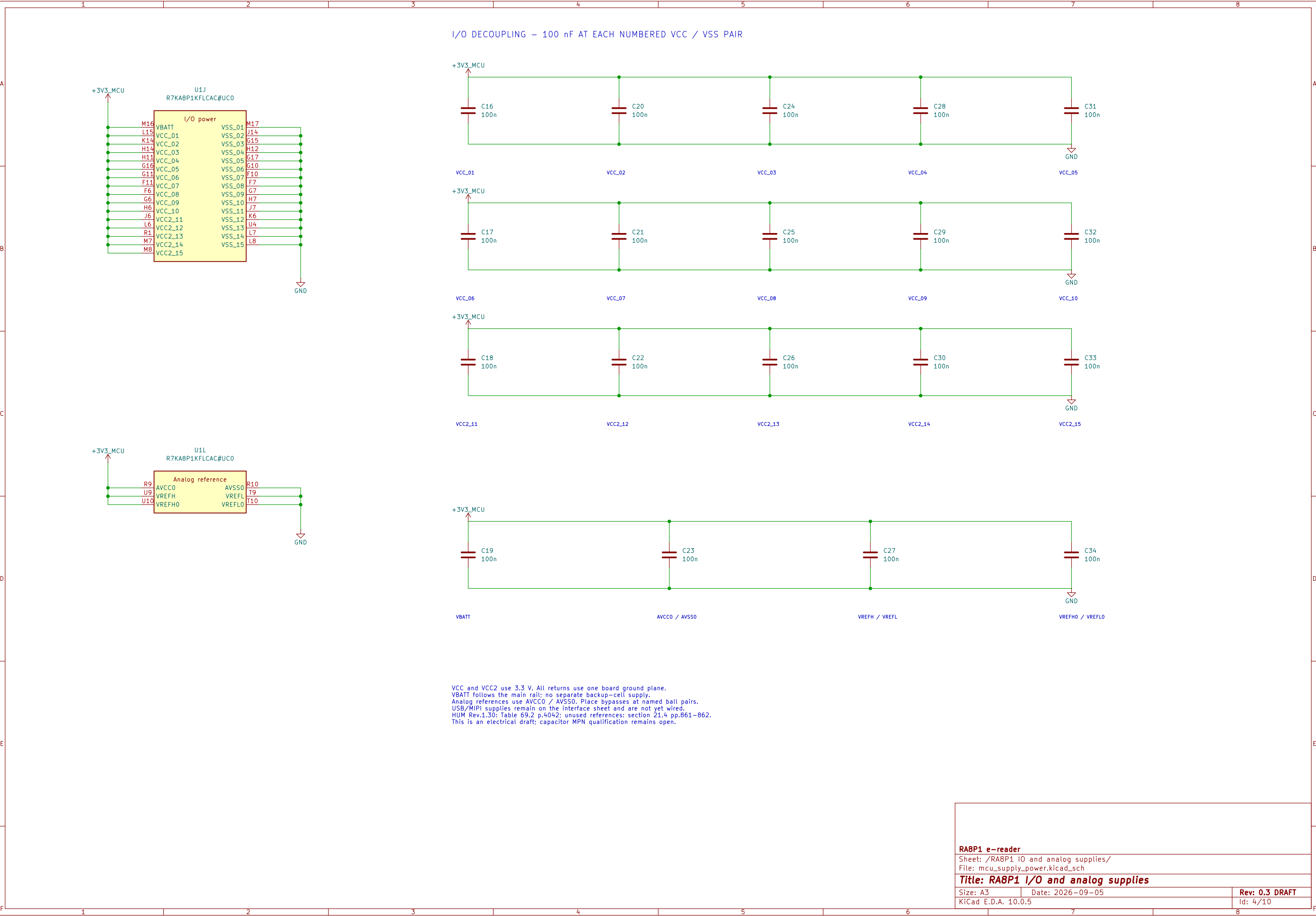


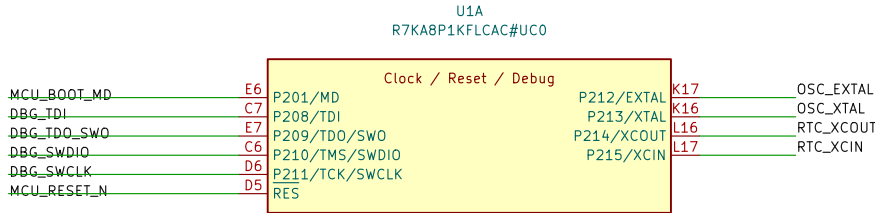
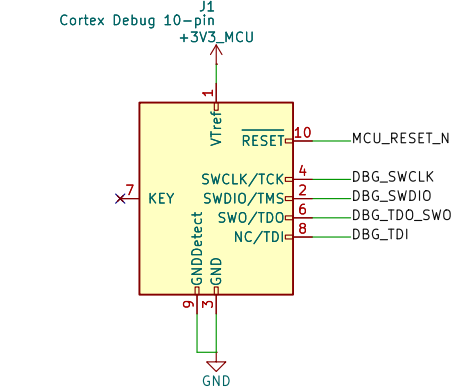
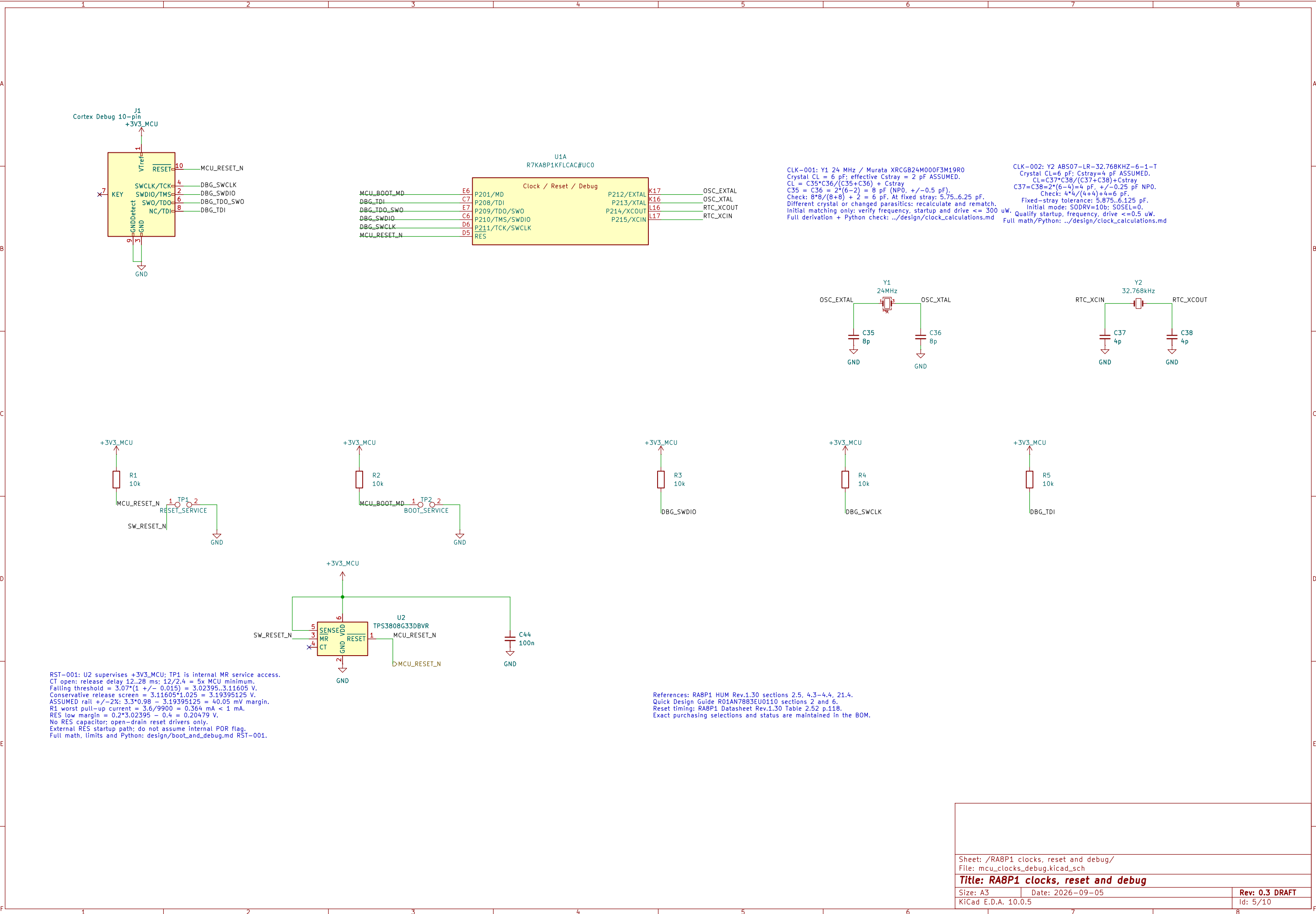
Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



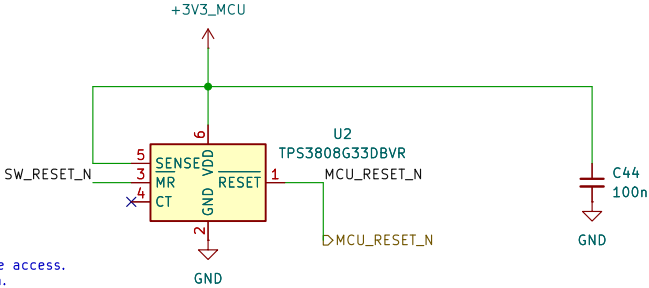
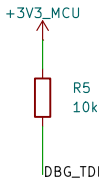
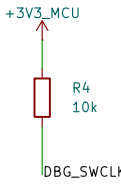
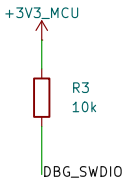
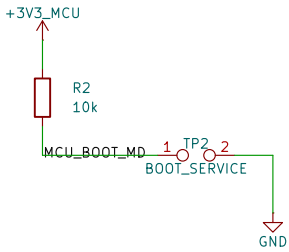
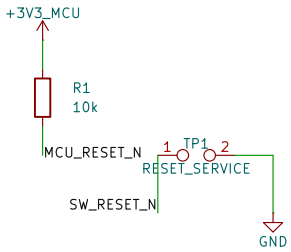
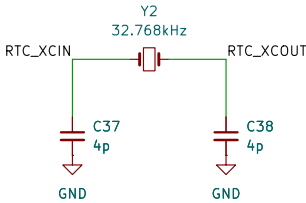
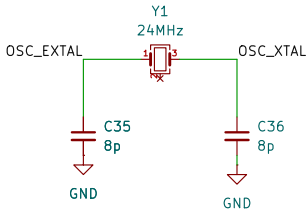






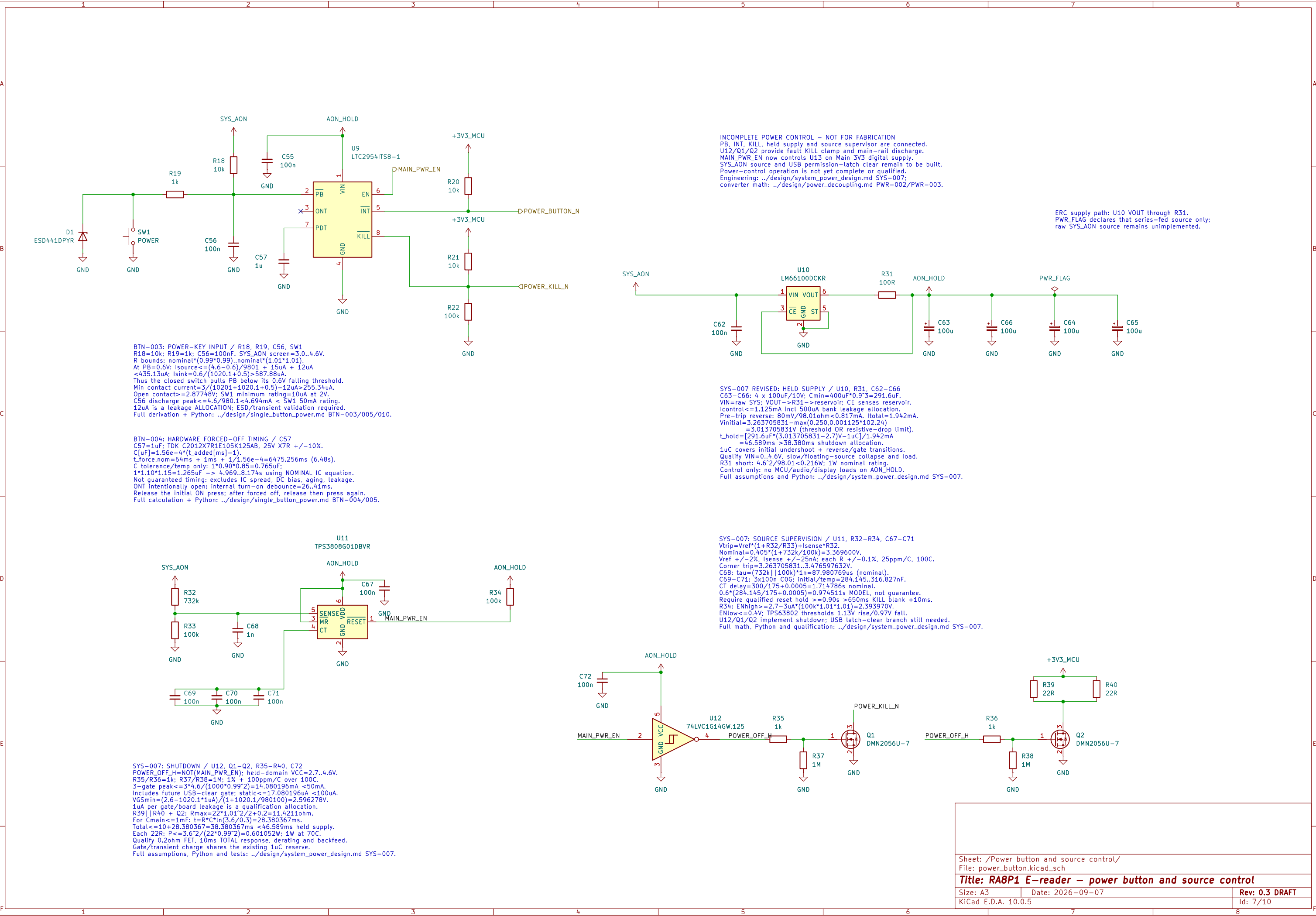
CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF; effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

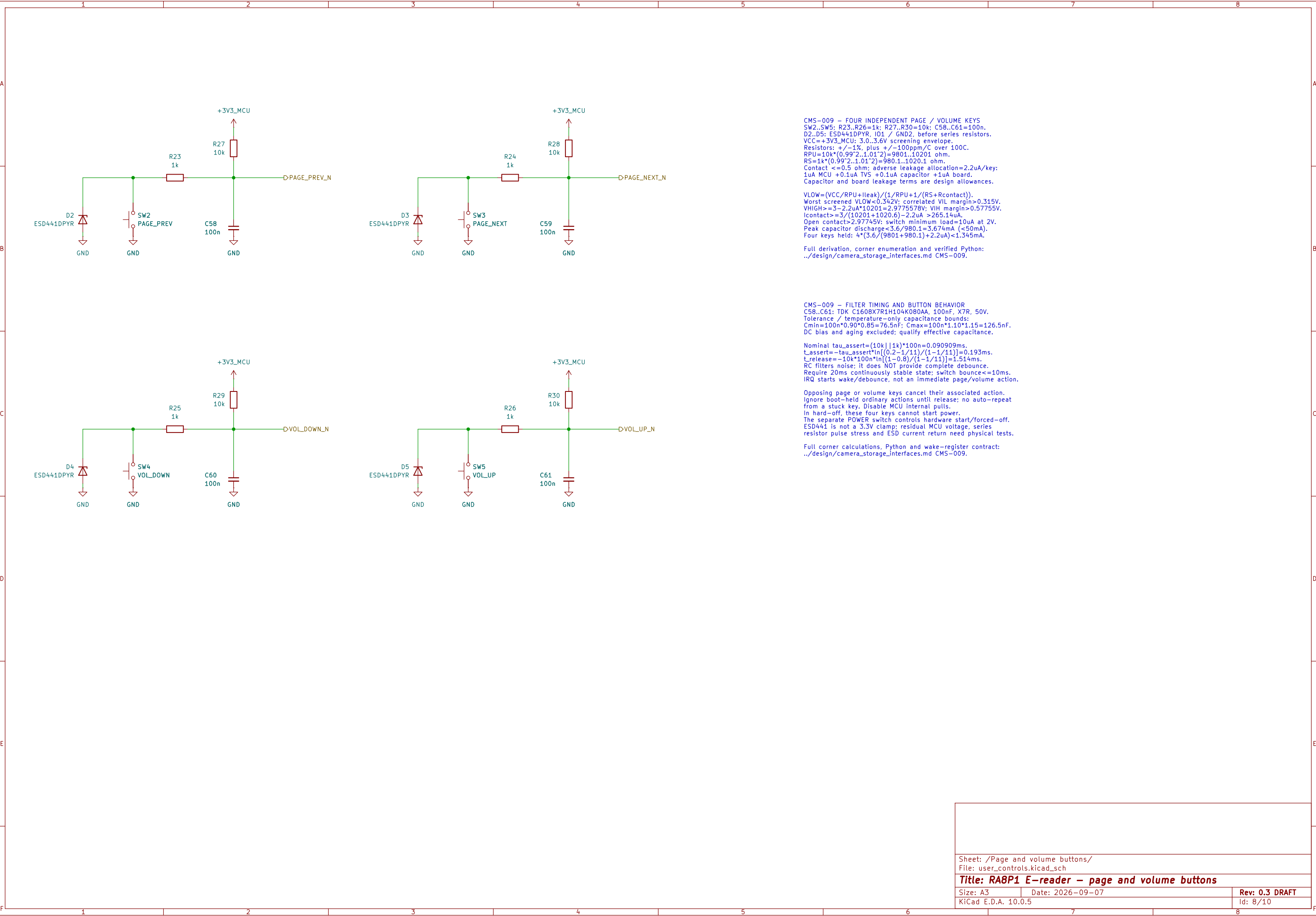
CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF; Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md



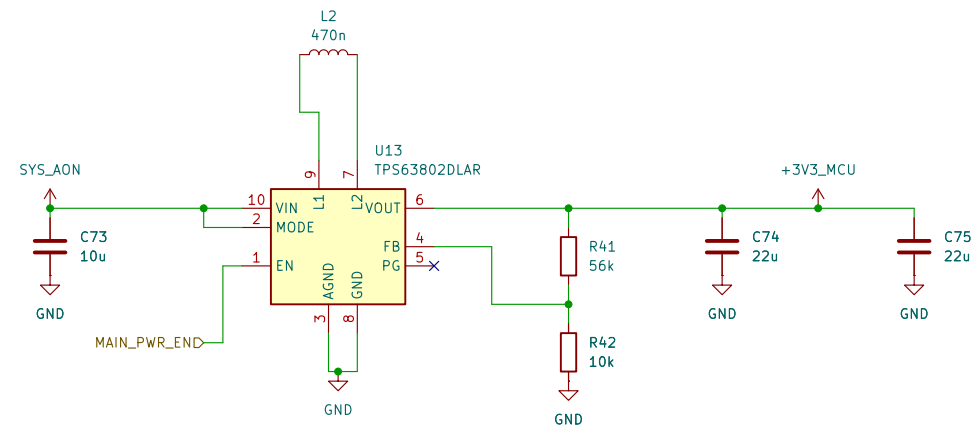
RST-001: U2 supervises +3V3_MCU; TP1 is internal MR service access.
CT open: release delay 12..28 ms; 12/2.4 = 5x MCU minimum.
Falling threshold = 3.07*(1 +/- 0.015) = 3.02395..3.11605 V.
Conservative release screen = 3.11605*1.025 = 3.19395125 V.
ASSUMED rail +/-2%: 3.3*0.98 - 3.19395125 = 40.05 mV margin.
R1 worst pull-up current = 3.6/9900 = 0.364 mA < 1 mA.
RES low margin = 0.2*3.02395 - 0.4 = 0.20479 V.
No RES capacitor; open-drain reset drivers only.
External RES startup path; do not assume internal POR flag.
Full math, limits and Python: design/boot_and_debug.md RST-001.

References: RABP1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RABP1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.





TPS63802 buck-boost; hardware-controlled enable; fixed PWM.
Engineering draft: upstream source and corner qualification remain open.



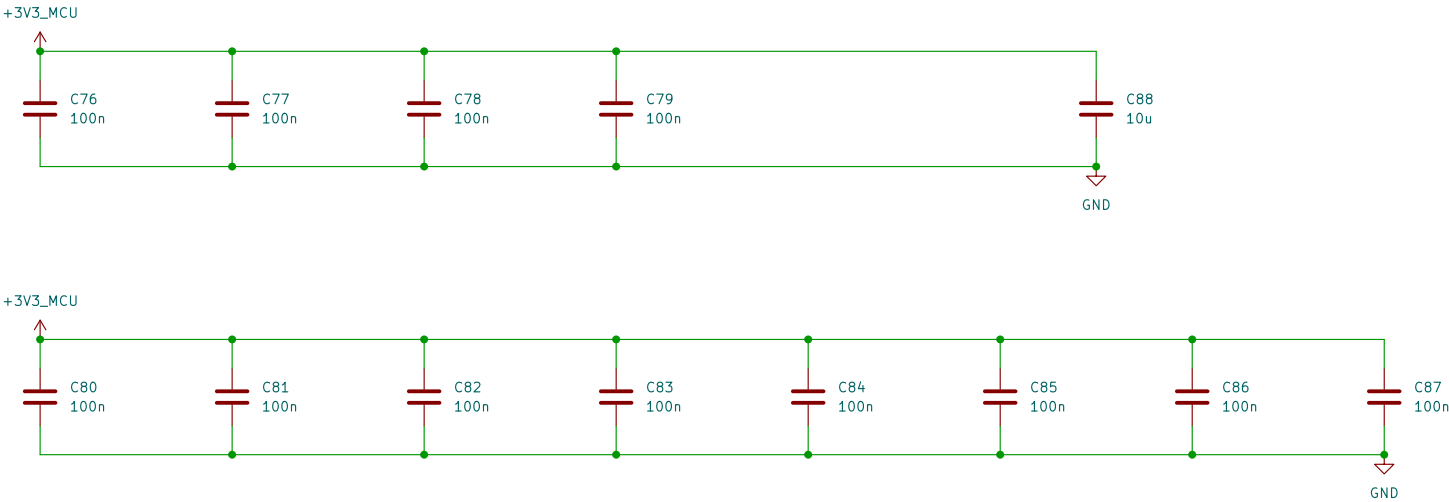
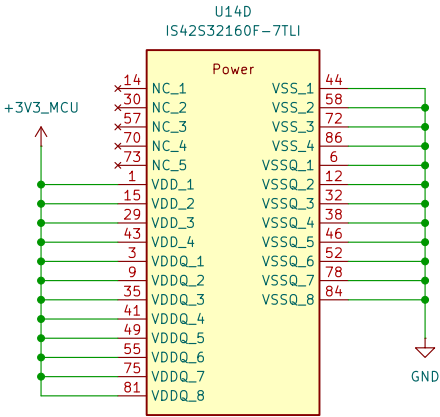
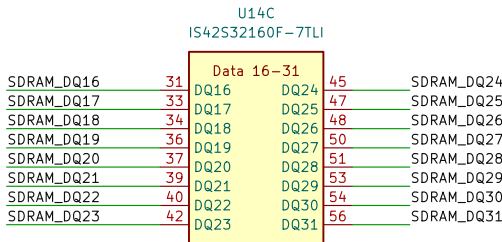
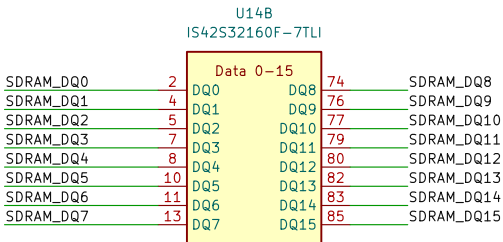
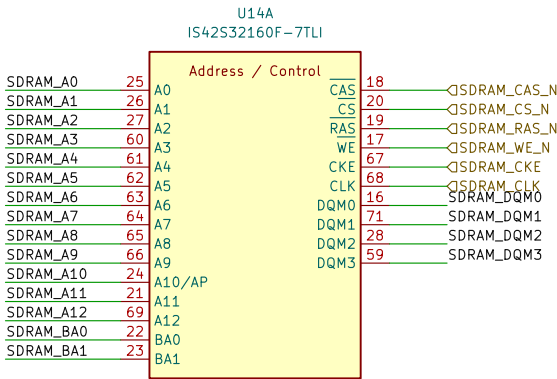
TI SLVSEU9D Table 7–1, sections 9.4/10.2; exact parts in native BOM.
Full derivation, primary sources and executable Python:
../design/power_decoupling.md PWR-002/PWR-003.

Pout=3.358485094*1.65=5.541500W.
VIN>=3.20V allocation: raw trip min=3.263705831V.
Allowed raw-to-VIN loss=3.263705831-3.20=63.705831mV.
At ASSUMED eta=85%: lin:Pout*(eta/3.20)=2.037316A.
Ploss=Pout*(1/eta-1)=0.977912W.
Reference rise=Ploss*81C/W=79.210859C: NOT enclosure prediction.
Verify hot current capability and junction temperature <125C with margin.

PWR-002 static release margin: radio 14.922537mV; MCU 48.092861mV.
Ripple, startup and transients must fit; no fabrication release.
Full sources, derivation and Python: `../design/power_decoupling.md` PWR-003;
source-loss shutdown: `../design/system_power_design.md` SYS-007.

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal interconnect phase | see CMS-010.
Pending: command pulls, MCU pin-role typing and timing qualification.

SDRAM_A[0..12]SDRAM_BA[0..1]SDRAM_DQM[0..3]SDRAM_DQ[0..31]



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

Title: RA8P1 E-reader - 64 MiB SDRAM

Size: A3 Date: 2026-09-08
KiCad E.D.A. 10.0.5

Rev: 0.3 DRAFT
Id: 10/10